

ALOR SETAR, Malaysia

WMO: 486030

Lat: 6.2005N

Lon: 100.4040E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 71.8	(c) 73.2	(d) 64.5	(e) 90.8	(f) 86.7	(g) 66.4	(h) 97.3	(i) 84.7	(j) 12.6	(k) 85.1	(l) 11.5	(m) 85.8	(n) 3.0	(o) 10	(p) 0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 15.5	(c) 96.7	(d) 78.6	(e) 95.0	(f) 79.6	(g) 93.4	(h) 80.2	(i) 84.8	(j) 91.7	(k) 84.2	(l) 91.1	(m) 83.5	(n) 90.1	(o) 7.7	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.7	170.8	90.4	82.4	168.8	90.1	81.0	161.3	88.4	48.9	92.8	48.3	90.9	47.4	90.6	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.3	(b) 11.3	(c) 9.9	(e) 69.4	(f) 98.4	(g) 1.8	(h) 1.9	(i) 68.1	(j) 99.7	(k) 67.0	(l) 100.8	(m) 66.0	(n) 101.9	(o) 64.7	(p) 103.3		
(a) 12.3	(b) 11.3	(c) 9.9	(e) 67.3	(f) 87.0	(g) 1.7	(h) 1.6	(i) 66.1	(j) 88.2	(k) 65.1	(l) 89.1	(m) 64.2	(n) 90.0	(o) 63.0	(p) 91.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	82.4	83.9	85.0	84.9	84.4	83.7	82.8	82.8	82.7	82.6	82.6	82.1
	DBStd	2.01	1.79	1.83	1.79	1.75	1.68	1.65	1.75	1.81	1.85	1.68	1.76	1.75
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12158	1004	950	1085	1048	1065	1010	1016	1016	980	1011	978	995
	CDD65	6683	539	530	620	598	600	560	551	551	530	546	528	530
	CDH74	71177	5528	5897	6986	6872	6964	6387	5892	5733	5337	5350	5026	5205
	CDH80	28974	2275	2847	3411	3107	2946	2580	2196	2147	1934	1903	1767	1861
Wind	WSAvg	3.9	4.5	4.9	4.3	3.9	3.9	3.5	3.4	3.6	3.6	3.6	3.3	4.1
Precipitation	PrecAvg	81.4	1.3	1.9	4.3	7.8	9.2	6.5	7.4	8.0	11.7	11.6	8.3	3.5
	PrecMax	99.6	10.2	12.7	7.9	19.0	17.3	15.3	12.8	13.9	25.0	17.1	13.9	9.9
	PrecMin	56.3	0.0	0.0	0.0	1.5	3.6	2.1	2.6	1.7	0.7	5.3	1.7	0.1
	PrecStd	11.7	2.0	2.7	2.1	4.3	3.4	3.1	3.1	3.3	4.9	3.7	3.7	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.0	97.2	98.9	97.0	94.9	93.3	91.8	93.2	93.2	93.1	93.2	93.1
		MCWB	76.8	76.7	77.8	82.1	83.6	83.6	82.7	80.6	82.2	82.6	80.5	79.7
	2%	DB	93.0	95.3	97.0	95.0	93.2	91.6	91.2	91.5	91.5	91.4	91.5	91.5
		MCWB	77.8	77.9	78.4	82.1	83.5	82.8	82.5	81.2	81.5	81.8	80.7	79.5
	5%	DB	91.4	93.6	95.3	93.3	91.6	91.0	89.8	90.0	89.9	89.8	89.9	89.9
		MCWB	77.7	77.9	78.9	82.3	82.8	82.5	81.8	81.0	80.9	81.1	80.5	79.1
	10%	DB	89.7	92.9	93.4	91.6	90.0	89.6	89.2	89.4	89.2	88.2	89.2	88.2
		MCWB	77.5	77.9	79.4	82.0	82.1	81.8	81.6	80.9	80.7	80.8	80.4	78.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.1	83.0	84.4	85.9	85.8	85.2	84.6	84.2	83.9	84.5	84.7	84.0
		MCDB	90.0	92.8	94.3	94.1	92.7	90.7	90.2	90.5	90.8	90.8	90.2	90.3
	2%	WB	81.4	81.4	82.9	84.5	84.5	84.1	83.6	82.8	82.5	83.1	83.1	82.3
		MCDB	88.7	91.3	92.1	92.5	91.4	90.1	89.3	89.0	89.2	89.4	89.1	88.6
	5%	WB	79.9	80.2	81.7	83.5	83.6	83.1	82.6	82.1	81.7	82.1	81.8	80.8
		MCDB	87.4	89.9	90.6	91.1	90.2	88.7	88.0	88.3	88.0	88.3	87.8	87.5
	10%	WB	78.8	79.4	80.6	82.6	82.7	82.3	81.9	81.2	80.8	81.0	80.5	79.6
		MCDB	86.2	88.7	89.4	89.8	88.7	87.6	87.2	87.0	86.8	86.9	86.4	86.0
Mean Daily Temperature Range	5% DB	MDBR	15.8	19.1	18.6	15.5	13.3	12.6	12.7	13.2	13.1	13.2	13.4	13.5
		MCDBR	18.2	20.5	21.5	17.0	14.5	13.7	13.8	14.4	14.5	14.8	15.2	15.9
		MCWBR	6.9	7.0	7.7	7.8	7.6	7.3	7.6	7.0	7.0	7.3	7.1	6.8
	5% WB	MCDBR	15.8	18.8	18.4	15.8	14.1	13.0	13.1	13.6	13.9	14.1	14.5	14.5
	MCWBR	7.9	8.2	8.4	8.4	8.0	7.6	7.9	7.6	7.3	7.9	8.0	7.7	
Clear-Sky Solar Irradiance	taub	0.438	0.459	0.470	0.475	0.460	0.472	0.474	0.476	0.468	0.461	0.442	0.437	
	taud	2.362	2.277	2.269	2.263	2.327	2.317	2.298	2.279	2.301	2.318	2.373	2.385	
	Ebn at Noon	279	276	273	267	265	258	259	263	269	272	276	277	
	Edh at Noon	40	44	45	45	41	40	42	43	43	42	39	38	
All-Sky Solar Radiation	RadAvg	1592	1858	1853	1806	1629	1577	1541	1561	1512	1470	1409	1378	
	RadStd	111	116	156	77	82	68	66	62	66	105	90	120	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.40	+0.25	N/A	+0.58	+0.40	+0.34	N/A	N/A	+169	+169

Nomenclature: See separate page